POLYNOMIALS 2

2.1 Introduction

In Class IX, you have studied polynomials in one variable and their degrees. Recall that if $p(x)$ is a polynomial in x, the highest power of x in $p(x)$ is called **the degree of the polynomial** $p(x)$. For example, $4x + 2$ is a polynomial in the variable x of degree 1, $2y^2 - 3y + 4$ is a polynomial in the variable y of degree 2, $5x^3 - 4x^2 + x - \sqrt{2}$ is a polynomial in the variable x of degree 3 and $7u^6 - \frac{3}{2}u^4 + 4u^2 + u - 8$ is a polynomial in the variable u of degree 6. Expressions like $\frac{1}{x-1}$, $\sqrt{x} + 2$, $\frac{1}{x^2 + 2x + 3}$ etc., are not polynomials.

A polynomial of degree 1 is called a **linear polynomial**. For example, $2x - 3$,

Algebraic expression

Combination of variables, constants
using basic arithmetic operators $(+, -, \times, \div)$

For example:

$2x$

$17x + y$

$\frac{1}{x} - 5$

$\frac{x}{3} + 1$

$\frac{1}{x} = x^{-1}$

5

Polynomial

It is an algebraic expression where the exponents on the variable must be a whole number.
(0, 1, 2, 3, ... ∞)

$\frac{x}{3} + 1, 2x$, 3^{-2}

$17x + 1$, 5

$\frac{17}{x} + 1$

Degree of polynomial

$$4x^{3} + 2x^{1} + 3x^{2} + 5x^{5}$$

$$3x^{3} + 20^{5} + x^{4}$$

$$3x^{3} + 3y^{4} + 12x^{2}$$

$$x^{2}y^{2} + 3x^{3} + 2x + 5$$

$$\frac{x^{9}}{y^{7}} + y^{5}$$

TYPES OF POLYNOMIAL

BASED ON THE NUMBER OF TERMS

No. of terms	Type	Example
1	Monomial	$5x$
2	Binomial	$3x+2$
3	Trinomial	$3x^2+3x^3+2$
4	Quadrinomial	$3x^4+2x^3+5x+3$

$3x+2x = 5x$

Like terms

- Same variable
- Same exponent

These terms must be unlike terms.

$4xy, -3xy$

Unlike $\begin{cases} 4xy, -3y \\ -4x^2y, -3xy \end{cases}$

TYPES OF POLYNOMIAL (Based on degree)

Degree	Types	Example	General form
0	Constant polynomial	7	a
1	Linear polynomial	$7x$	ax
2	Quadratic polynomial	x^2+2	ax^2+bx+c
3	Cubic polynomial	x^3+x^2+1	ax^3+bx^2+cx+d

$a \neq 0$

Not defined

Zero polynomial — 0 → 0

a, b, c, d are integers

$f(x) = x+2$

$y = x+2$

(x, y)

put $y = 0$

$0 = x+2$

$x = -2$

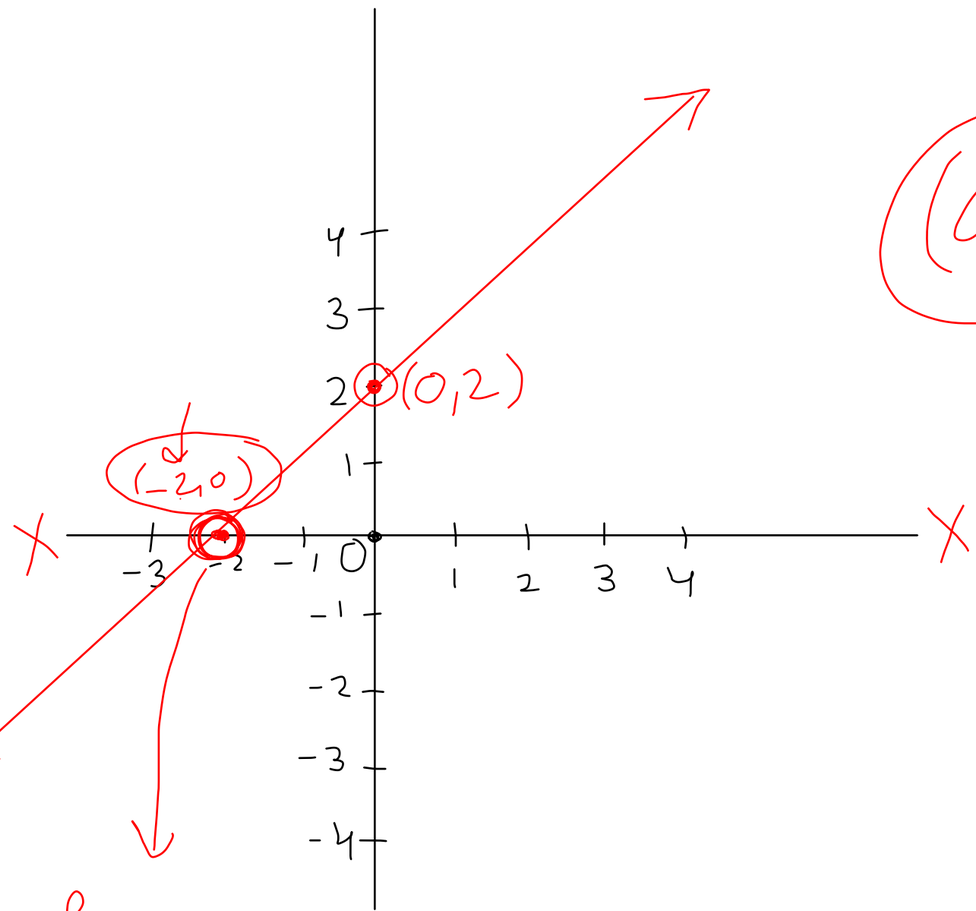
$(-2, 0)$

$(0, 2)$

put $x = 0$

$y = 0+2$

$y = 2$



for $x = -2$, $p(x) = 0$

for a polynomial $p(x)$ of degree n, the graph of $y = p(x)$ intersects the x-axis at atmost 'n' points.

2.2 Geometrical Meaning of the Zeroes of a Polynomial

17/03/26

You know that a real number k is a zero of the polynomial $p(x)$ if $p(k) = 0$. But why are the zeroes of a polynomial so important? To answer this, first we will see the **geometrical** representations of linear and quadratic polynomials and the geometrical meaning of their zeroes.

Consider first a linear polynomial $ax + b$, $a \neq 0$. You have studied in Class IX that the graph of $y = ax + b$ is a straight line. For example, the graph of $y = 2x + 3$ is a straight line passing through the points $(-2, -1)$ and $(2, 7)$.

x	-2	2
$y = 2x + 3$	-1	7

From Fig. 2.1, you can see that the graph of $y = 2x + 3$ intersects the x-axis mid-way between $x = -1$ and $x = -2$, that is, at the point $\left(-\frac{3}{2}, 0\right)$. You also know that the zero of $2x + 3$ is $-\frac{3}{2}$. Thus, the zero of the polynomial $2x + 3$ is the x-coordinate of the point where the graph of $y = 2x + 3$ intersects the x-axis.

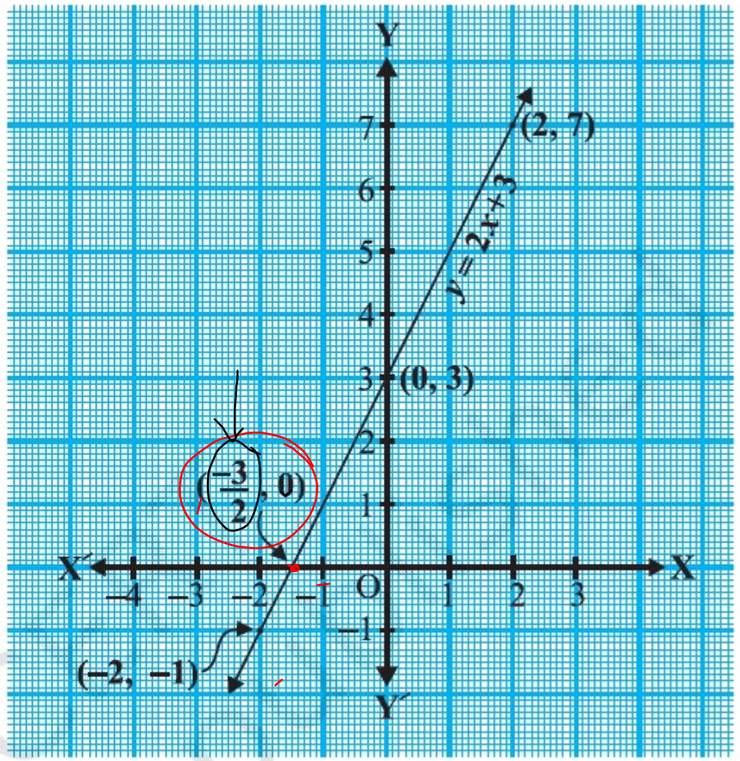
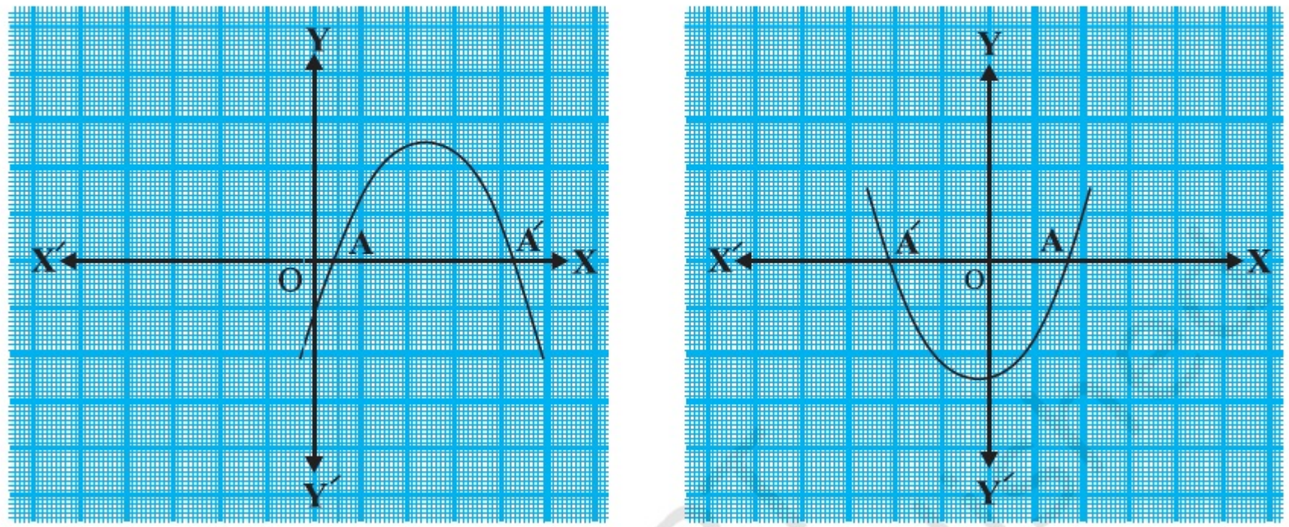


Fig. 2.1

Case (i) : Here, the graph cuts x-axis at two distinct points A and A′.

The x-coordinates of A and A′ are the **two zeroes** of the quadratic polynomial $ax^2 + bx + c$ in this case (see Fig. 2.3).



(i) (ii)

Fig. 2.3

Case (ii) : Here, the graph cuts the x-axis at exactly one point, i.e., at two coincident points. So, the two points A and A′ of Case (i) coincide here to become one point A (see Fig. 2.4).

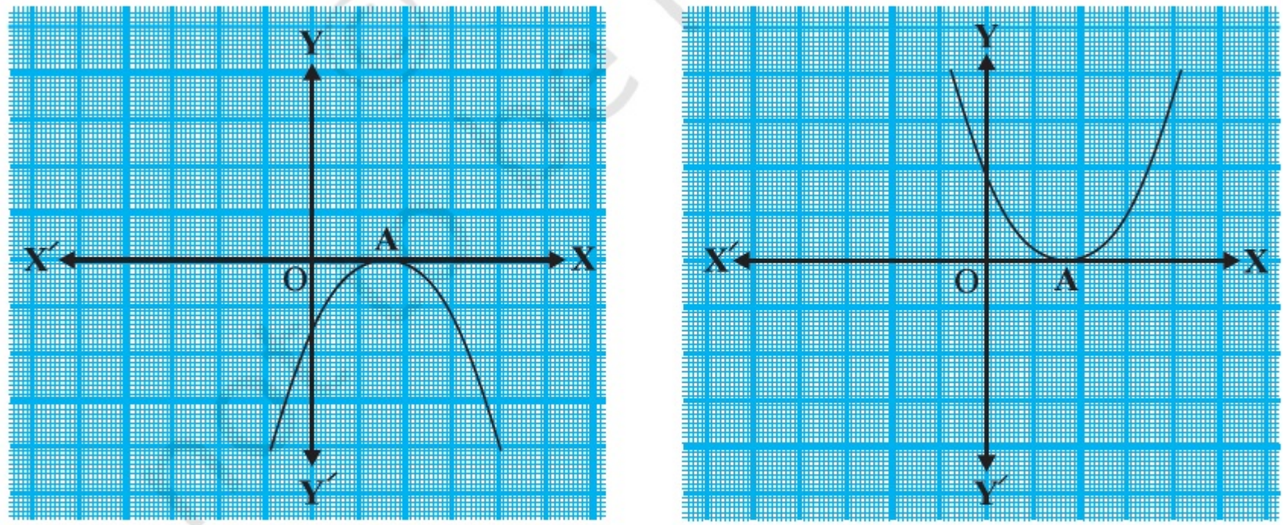


Fig. 2.4

The x-coordinate of A is the **only zero** for the quadratic polynomial $ax^2 + bx + c$ in this case.

Case (iii) : Here, the graph is either completely above the x-axis or completely below the x-axis. So, it does not cut the x-axis at any point (see Fig. 2.5).

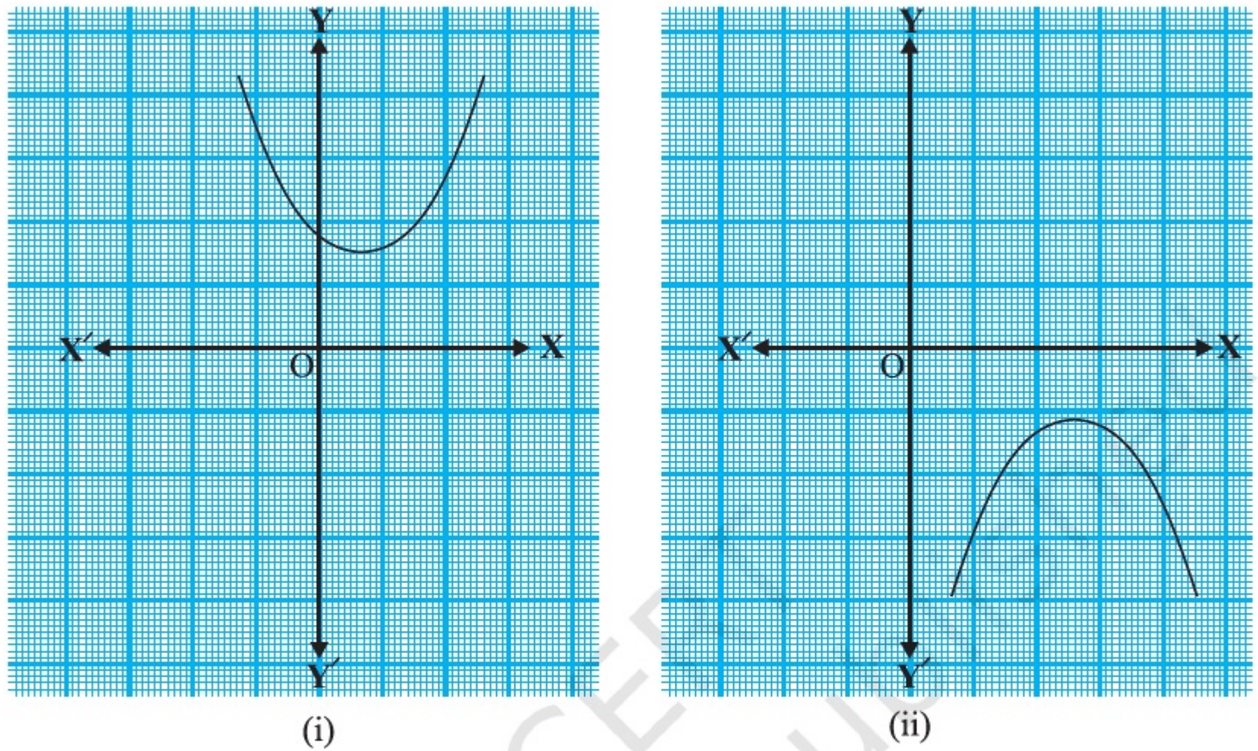


Fig. 2.5

So, the quadratic polynomial $ax^2 + bx + c$ has **no zero** in this case.

So, you can see geometrically that a quadratic polynomial can have either two distinct zeroes or two equal zeroes (i.e., one zero), or no zero. This also means that a polynomial of degree 2 has atmost two zeroes.

Now, what do you expect the geometrical meaning of the zeroes of a cubic polynomial to be? Let us find out. Consider the cubic polynomial $x^3 - 4x$. To see what the graph of $y = x^3 - 4x$ looks like, let us list a few values of y corresponding to a few values for x as shown in Table 2.2.

Table 2.2

x	-2	-1	0	1	2
$y = x^3 - 4x$	0	3	0	-3	0

Locating the points of the table on a graph paper and drawing the graph, we see that the graph of $y = x^3 - 4x$ actually looks like the one given in Fig. 2.6.

We see from the table above that -2, 0 and 2 are zeroes of the cubic polynomial $x^3 - 4x$. Observe that -2, 0 and 2 are, in fact, the x-coordinates of the only points where the graph of $y = x^3 - 4x$ intersects the x-axis. Since the curve meets the x-axis in only these 3 points, their x-coordinates are the only zeroes of the polynomial.

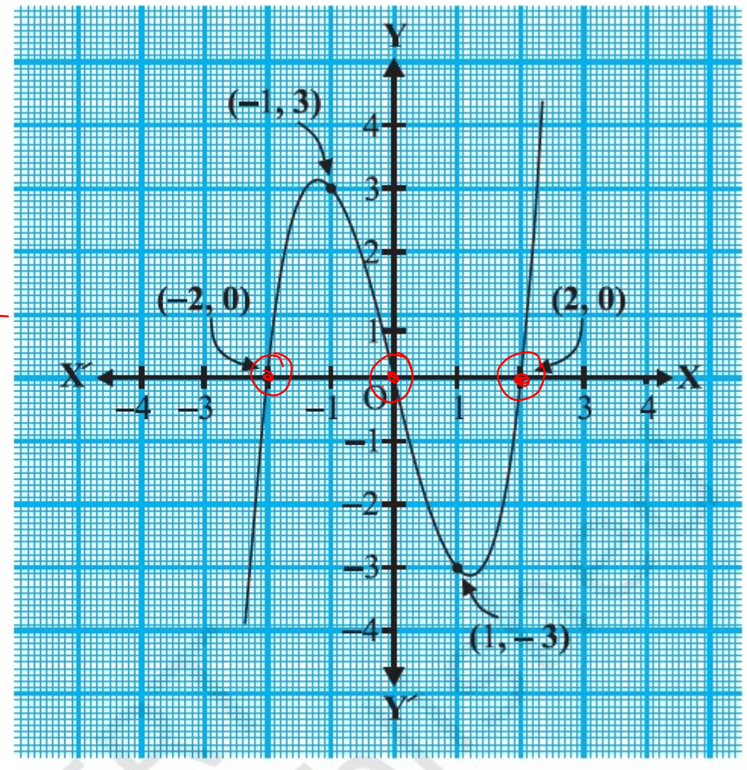
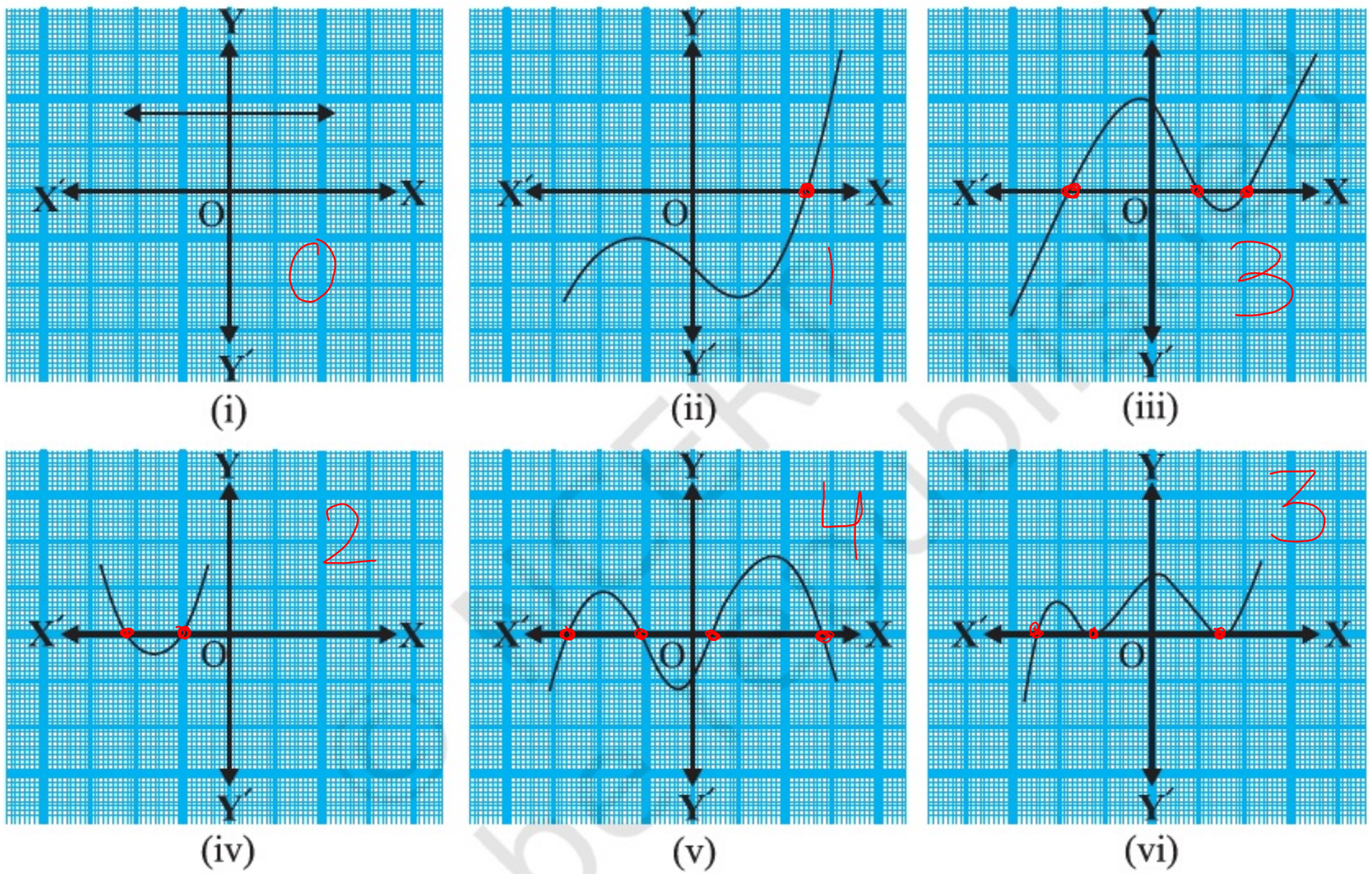


Fig. 2.6

Let us take a few more examples. Consider the cubic polynomials x^3 and $x^3 - x^2$. We draw the graphs of $y = x^3$ and $y = x^3 - x^2$ in Fig. 2.7 and Fig. 2.8 respectively.

EXERCISE 2.1

1. The graphs of $y = p(x)$ are given in Fig. 2.10 below, for some polynomials $p(x)$. Find the number of zeroes of $p(x)$, in each case.



(i) (ii) (iii)

(iv) (v) (vi)

Fig. 2.10

$p(x) = x^2 - 4 \Rightarrow y = x^2 - 4$

x	0	1	-1	2	-2	3	-3
y	-4	-3	-3	0	0	5	5

$x = 0, 1, -1, 2, -2,$

Put $x = 0$

$y = 0^2 - 4$

$y = -4$

Put $x = 1$

$y = 1^2 - 4$

$y = 1 - 4$

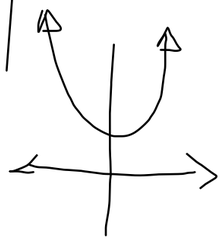
$= -3$

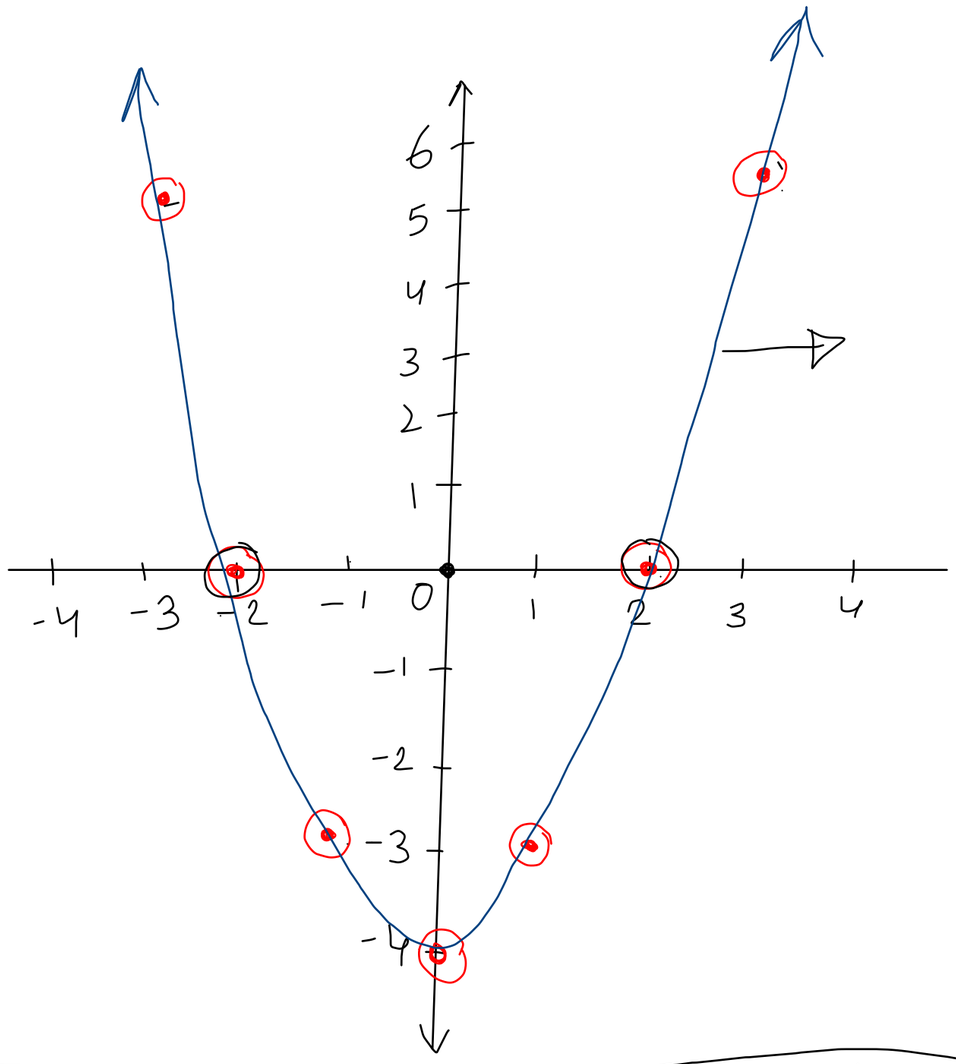
Put $x = (-1)$

$y = (-1)^2 - 4$

$= 1 - 4$

$= -3$





HW Draw the graph of the polynomial $p(x) = x^2 + 2$ & find the number of zeros.

19/3/2026

$$p(x) = x^2 - 4$$

$x = 0, 1, -1, 2, -2, 3, -3$

x	0	1	-1	2	-2	3	-3
y	-4	-3	-3	0	0	5	5

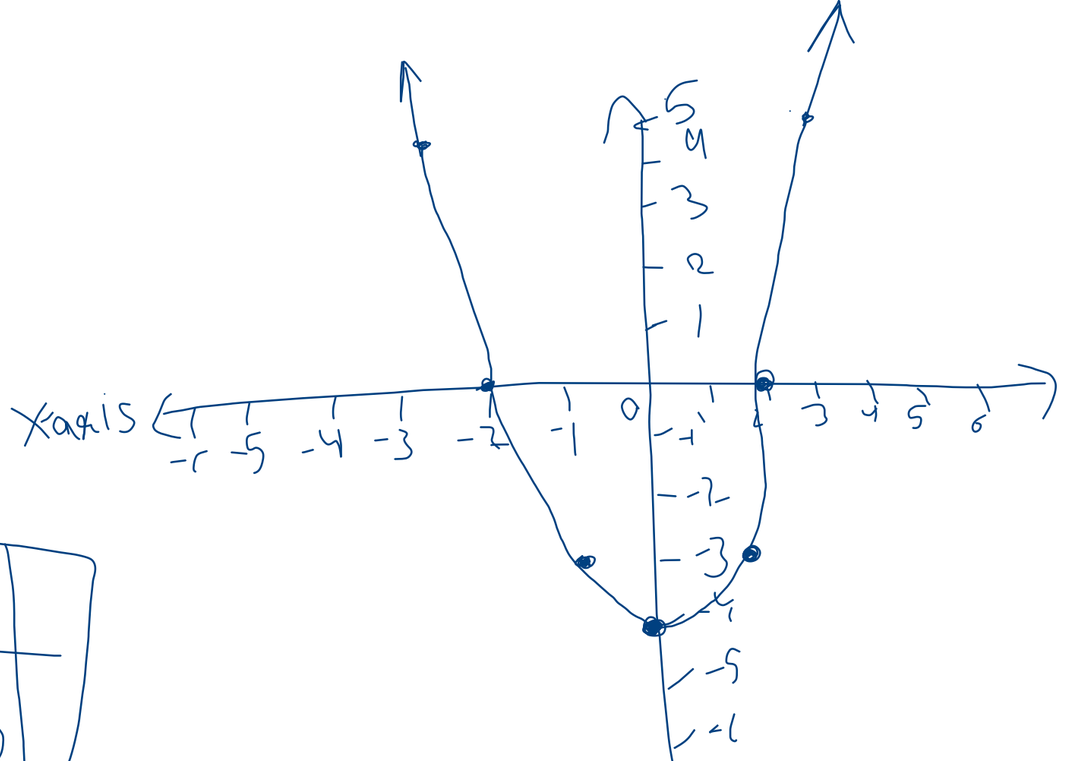
$y = 0^2 - 4$
$= -4$

$y = 1^2 - 4$
$= -3$

$y = (-1)^2 - 4$
$= 1 - 4$
$= -3$

$y = 2^2 - 4$
$= 4 - 4$
$= 0$

$y = 3^2 - 4$
$= 9 - 4$
$= 5$



2.3 Relationship between Zeroes and Coefficients of a Polynomial

In general, if α* and β* are the zeroes of the quadratic polynomial $p(x) = ax^2 + bx + c$, $a \neq 0$, then you know that $x - \alpha$ and $x - \beta$ are the factors of $p(x)$. Therefore,

$$ax^2 + bx + c = k(x - \alpha)(x - \beta), \text{ where } k \text{ is a constant}$$
$$= k[x^2 - (\alpha + \beta)x + \alpha\beta]$$
$$= kx^2 - k(\alpha + \beta)x + k\alpha\beta$$

Comparing the coefficients of x^2, x and constant terms on both the sides, we get

$a = k$, $b = -k(\alpha + \beta)$ and $c = k\alpha\beta$.

This gives

$$\alpha + \beta = \frac{-b}{a},$$

$$\alpha\beta = \frac{c}{a}$$

* α, β are Greek letters pronounced as 'alpha' and 'beta' respectively. We will use later one more letter 'γ' pronounced as 'gamma'.

18/03/26

MIDDLE TERM SPLITTING

$$p(x) = x^2 + 7x + 10 = 0$$

$$x^2 + 7x + 10$$

$$x^2 + 5x + 2x + 10$$

$$x(x+5) + 2(x+5)$$

$$\Rightarrow (x+5)(x+2) = 0$$

Either $x+5 = 0$

$x = -5$

OR $x+2 = 0$

OR $x = -2$

$5 \times 2 = 10$

$5 + 2 = 7$

Types of poly.	No. of zeroes	General form	Zeros
Linear pol^n	1	$ax+b$	α
Quadratic pol^n	2	ax^2+bx+c	α, β
Cubic pol^n	3	ax^3+bx^2+cx+d	α, β, γ
Constant	0	a	

$a \neq 0$

$\alpha \rightarrow$ Alpha

$\beta \rightarrow$ Beta

$\gamma \rightarrow$ Gamma

① Linear polynomial

$p(x) = ax + b$

$\alpha = -\frac{b}{a}$

$\alpha = -\frac{\text{constant term}}{\text{coeff. of } x}$

$p(x) = 2x + 5$

$2x + 5 = 0$

$2x = -5$

$x = -\frac{5}{2}$

$\alpha = -\frac{5}{2}$

$ax + b = 0$

$ax = -b$

$x = -\frac{b}{a}$

ⓐ

2

Example 2 : Find the zeroes of the quadratic polynomial $x^2 + 7x + 10$, and verify the relationship between the zeroes and the coefficients.

$x^2+7x+10=0$

$x^2+5x+2x+10=0$

$x(x+5)+2(x+5)=0$

$(x+5)(x+2)=0$

Either $x+5=0$ OR $x+2=0$

$x=-5$ OR $x=-2$

$\alpha=-5$

$\beta=-2$

$a=1$

$b=7$

$c=10$

Relation 1

$\alpha+\beta=-5+(-2)$

$=-7$

$\frac{-b}{a}=\frac{-7}{1}=-7$

$\alpha+\beta=\frac{-b}{a}$

Relation 2

$\alpha\beta=\frac{c}{a}$

$\alpha\beta=(-5)\times(-2)$

$=10$

$\frac{c}{a}=\frac{10}{1}$

$=10$

☆ For a quadratic polynomial $p(x) = ax^2 + bx + c$

zeroes will be α & β.

① $\alpha + \beta = \frac{-b}{a}$; Sum of zeroes $= \frac{-\text{coeff of } x}{\text{coeff of } x^2}$

② $\alpha\beta = \frac{c}{a}$; Product of zeroes $= \frac{\text{constant term}}{\text{coeff of } x^2}$

$a \rightarrow$ coefficient of x^2

$b \rightarrow$ coefficient of x

$c \rightarrow$ constant term

EXERCISE 2.2

1. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

(i) $x^2 - 2x - 8$ (ii) $4s^2 - 4s + 1$ (iii) $6x^2 - 3 - 7x$

(iv) $4u^2 + 8u$ (v) $t^2 - 15$ (vi) $3x^2 - x - 4$

$\alpha + \beta = \frac{-b}{a}$ & $\alpha\beta = \frac{c}{a}$

(i) $p(x) = x^2 - 2x - 8$

To find the zeroes

put $p(x) = 0$

$x^2 - 2x - 8 = 0$

$x^2 - 4x + 2x - 8 = 0$

$x(x-4) + 2(x-4) = 0$

$(x-4)(x+2) = 0$

Either $x - 4 = 0$ OR $x + 2 = 0$

$x = 4$ OR $x = -2$

$\alpha = 4$; $a = 1$

$\beta = -2$; $b = -2$

$c = -8$

Verification: (1) $\alpha + \beta = \frac{-b}{a}$ ✓

$\alpha + \beta = 4 + (-2)$ & $\frac{-b}{a} = \frac{-(-2)}{1}$

$= 2$ $= 2$

Verified

(2) $\alpha\beta = \frac{c}{a}$

$\alpha\beta = 4(-2)$

$= -8$

$\frac{c}{a} = \frac{-8}{1} = -8$

Verified

EXERCISE 2.2

1. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

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(iv) $4u^2 + 8u$ (v) $t^2 - 15$ (vi) $3x^2 - x - 4$

(iii) $p(x) = 6x^2 - 3 - 7x$

$= 6x^2 - 7x - 3$

to find 0, let $p(x) = 0$

$\Rightarrow 6x^2 - 7x - 3 = 0$

$\Rightarrow 6x^2 - 9x + 2x - 3 = 0$

$\Rightarrow 3x(2x - 3) + 1(2x - 3) = 0$

$\Rightarrow (2x - 3)(3x + 1) = 0$

Either $2x - 3 = 0$ or $3x + 1 = 0$

$2x - 3 = 0$

$x = \frac{3}{2}$

$3x = -1$

$x = \frac{-1}{3}$

$a = 6$

$b = -7$

$\alpha = \frac{3}{2}$

$\beta = \frac{-1}{3}$

(1) $\alpha + \beta = \frac{-b}{a}$

$\frac{3}{2} + \left(\frac{-1}{3}\right) = \alpha + \beta$

(2) $\alpha\beta = \frac{c}{a}$

$\frac{9}{6} + \left(\frac{-2}{6}\right) = \alpha + \beta$

$\frac{-b}{a} = \frac{-(-7)}{6} \Rightarrow \frac{7}{6}$

$\frac{7}{6} = \alpha + \beta$

EXERCISE 2.2

1. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

(i) $x^2 - 2x - 8$ (ii) $4s^2 - 4s + 1$ (iii) $6x^2 - 3 - 7x$

(iv) $4u^2 + 8u$ (v) $t^2 - 15$ (vi) $3x^2 - x - 4$

$p(u) = 4u^2 + 8u$

$4u^2 + 8u = 0$

$4u(u+2) = 0$

either $4u = 0$ or $(u+2) = 0$

$u = \frac{0}{4}$ $u = -2$

$= 0$

$a = 4$, $b = 8$, $c = 0$; $\alpha = 0$, $\beta = -2$

(1) $\alpha + \beta = 0 + (-2)$

$= -2$

$\frac{b}{a} = \frac{-8}{4} = -2$

Hence verified

(2) $\alpha\beta = 0 \times (-2)$

$= 0$

$\frac{c}{a} = \frac{0}{4} = 0$

Hence Verified ☺

EXERCISE 2.2

1. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

(i) $x^2 - 2x - 8$ (ii) $4s^2 - 4s + 1$ (iii) $6x^2 - 3 - 7x$

(iv) $4u^2 + 8u$ (v) $t^2 - 15$ (vi) $3x^2 - x - 4$

(v) $t^2 - 15 = 0$

$\Rightarrow t^2 - (\sqrt{15})^2 = 0$

Using the identity $a^2 - b^2 = (a+b)(a-b)$

$\Rightarrow (t+\sqrt{15})(t-\sqrt{15}) = 0$

$= t + \sqrt{15}$ | $= t - \sqrt{15}$

$t + \sqrt{15} = 0$ | $t = \sqrt{15}$

$t = -\sqrt{15}$

$\alpha = -\sqrt{15}$ | $\beta = \sqrt{15}$

Sum of zeroes $= \alpha + \beta = \frac{-b}{a}$

$a = 1, b = 0, C = -15$

L.H.S $= \alpha + \beta$

$= -\sqrt{15} + \sqrt{15}$

$= 0$

R.H.S $= \frac{-b}{a}$

$= \frac{0}{1}$

$= 0$

L.H.S = R.H.S

Product of zeroes $= \alpha\beta = \frac{c}{a}$

L.H.S $= \alpha\beta$

$= -\sqrt{15} \times \sqrt{15}$

$= -15$

L.H.S = R.H.S

R.H.S $= \frac{c}{a}$

$= \frac{-15}{1}$

$= -15$

Hence proved

EXERCISE 2.2

1. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

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(iv) $4u^2 + 8u$ (v) $t^2 - 15$ (vi) $3x^2 - x - 4$

(vi) $p(x) = 3x^2 - x - 4$

$3x^2 - x - 4 = 0$

$3x^2 - 4x + 3x - 4 = 0$

$x(3x - 4) + 1(3x - 4) = 0$

$(x + 1)(3x - 4) = 0$

either $x + 1 = 0$ or $3x - 4 = 0$

$x = -1$ or $x = \frac{4}{3}$

$\alpha = -1$ $\quad a = 3$

$\beta = \frac{4}{3}$ $\quad b = -1$

$c = -4$

i) $\alpha + \beta = \frac{-b}{a}$

$\alpha + \beta = -1 + \frac{4}{3}$ & $\frac{-b}{a} = \frac{-(-1)}{3} = \frac{1}{3}$

$= \frac{-3 + 4}{3}$

$= \frac{1}{3}$

Verified

ii) $\alpha \times \beta = \frac{c}{a}$

$\alpha \times \beta = -1 \times \frac{4}{3}$

$= \frac{-4}{3}$

$\frac{c}{a} = \frac{-4}{3}$

Verified

24/03/26

$p(x) = x^2 - 2x - 8$

$x^2 - 2x - 8 = 0$

$x^2 - 4x + 2x - 8 = 0$

$x(x-4) + 2(x-4) = 0$

$(x-4)(x+2) = 0$

$x - 4 = 0$ OR $x + 2 = 0$

$x = 4$ $\quad x = -2$

$\times 2$ | K

$p(x) = 2x^2 - 4x - 16$

$ax^2 + bx + c = 0$

$\frac{ax^2 + bx + c}{a} = \frac{0}{a}$

$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

$x^2 - \left(-\frac{b}{a}\right)x + \frac{c}{a} = 0$

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

Formula for quadratic polynomial

when α, β is given

$$K\left[x^2-(\alpha+\beta)x+\alpha\beta\right]$$

$$K\left[x^2-(\text{Sum of zeroes})x+(\text{Product of zeroes})\right]$$

✓

2. Find a quadratic polynomial each with the given numbers as the sum and product of its zeroes respectively.

(i) $\frac{1}{4}, -1$

(ii) $\sqrt{2}, \frac{1}{3}$

(iii) $0, \sqrt{5}$

(iv) $1, 1$

(v) $-\frac{1}{4}, \frac{1}{4}$

(vi) $4, 1$

$k\left[x^2 - \left(\frac{1}{4}\right)x + (-1)\right]$

$k\left[x^2 - \frac{x}{4} - 1\right]$

$k = 4$

$4\left(x^2 - \frac{x}{4} - 1\right)$

$= 4x^2 - x - 4$

(ii) $x^2 - (\text{sum of zeroes})x + \text{product}$

$k\left[x^2 - \sqrt{2}x + \frac{1}{3}\right]$

$k = 3$

$3\left(x^2 - \sqrt{2}x + \frac{1}{3}\right)$

$3x^2 - 3\sqrt{2}x + 1 = p(x)$

2. Find a quadratic polynomial each with the given numbers as the sum and product of its zeroes respectively.

(i) $\frac{1}{4}, -1$

(ii) $\sqrt{2}, \frac{1}{3}$

(iii) $0, \sqrt{5}$

(iv) $1, 1$

(v) $-\frac{1}{4}, \frac{1}{4}$

(vi) $4, 1$

(ii) $\alpha+\beta = \sqrt{2}$

$\alpha\beta = \frac{1}{3}$

$p(x) = k\left[x^2 - (\alpha+\beta)x + \alpha\beta\right]$

$\Rightarrow k\left[x^2 - \sqrt{2}x + \frac{1}{3}\right]$

$k=3$

$3\left(x^2 - \sqrt{2}x + \frac{1}{3}\right)$

$\Rightarrow \boxed{3x^2 - 3\sqrt{2}x + 1}$ Ans

1. Find the zeros of each of the following quadratic polynomials and verify the relationship between the zeros and their coefficients:

$$p(x) = x^2 + 2\sqrt{2}x - 6$$

$a = 1$
$b = 2\sqrt{2}$
$c = -6$

To find the zeroes of the polynomial, $p(x) = 0$

$$x^2 + 2\sqrt{2}x - 6 = 0$$

$3 \times \sqrt{2} = 3\sqrt{2}$, $\sqrt{2} \times \sqrt{2} = 2$

$$x^2 + 3\sqrt{2}x - \sqrt{2}x - 6 = 0$$

$$x(x + 3\sqrt{2}) - \sqrt{2}(x + 3\sqrt{2}) = 0$$

$$(x + 3\sqrt{2})(x - \sqrt{2}) = 0$$

$x + 3\sqrt{2} = 0$ OR $x - \sqrt{2} = 0$

$x = -3\sqrt{2}$ OR $x = \sqrt{2}$

$\alpha = -3\sqrt{2}$
$\beta = \sqrt{2}$

$3\sqrt{2}$ $\quad$ $-\sqrt{2}$

$\bigcirc \times \bigcirc = -6$

$\bigcirc + \bigcirc = 2\sqrt{2}$

Verification

(i) $\alpha + \beta = -\frac{b}{a}$

$-3\sqrt{2} + \sqrt{2}$ $\quad$ $-\frac{2\sqrt{2}}{1}$

$\Rightarrow -2\sqrt{2}$ $\quad$ $= -2\sqrt{2}$

Verified

(ii) $\alpha\beta = \frac{c}{a}$

$-3\sqrt{2} \times \sqrt{2}$ $\quad$ $\frac{-6}{1}$

$\Rightarrow -3 \times 2$ $\quad$ $= -6$

$= -6$

Verified.

2. For each of the following, find a quadratic polynomial whose sum and product respectively of the zeros are as given. Also, find the zeros of these polynomials by factorization.

(i) -8/3 , 4/3

Sum $-\frac{8}{3}$, Product $\frac{4}{3}$

$k(x^2 - (\text{sum of zeros})x + \text{product of zeros})$

$k(x^2 - (-\frac{8}{3})x + \frac{4}{3})$

$k(x^2 + \frac{8}{3}x + \frac{4}{3})$

$k = 3$

$p(x) = 3x^2 + 8x + 4$

$3x^2 + 8x + 4 = 0$

$3x^2 + 6x + 2x + 4 = 0$

$3x(x+2) + 2(x+2) = 0$

$(x+2)(3x+2) = 0$

$x + 2 = 0$ or $3x + 2 = 0$

$x = -2$ ✓ or $3x = -2$

$x = \frac{-2}{3}$ ✓

4. If α and β are the zeros of the quadratic polynomial $p(y) = 5y^2 - 7y + 1$, find the value of $1/\alpha + 1/\beta$.

To find $\frac{1}{\alpha} + \frac{1}{\beta}$

$a = 5$

$b = -7$

$c = 1$

S.

-7

$\alpha + \beta = -\frac{b}{a}$ *

$\alpha + \beta = \frac{-(-7)}{5} = \frac{7}{5}$

$\alpha\beta = \frac{1}{5}$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} = \frac{\alpha + \beta}{\alpha\beta}$

$= \frac{\frac{7}{5}}{\frac{1}{5}}$

$= 7$

3. If α and β are the zeros of the quadratic polynomial $f(x) = x^2 - 5x + 4$, find the value of $1/\alpha + 1/\beta - 2\alpha\beta$.

To find: $\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$

$\alpha + \beta = -\frac{b}{a}$

$= -(-5)$

$= 5$

$\alpha + \beta = 5$ ✓

$\alpha\beta = 4$ ✓

$\frac{\alpha+\beta}{\alpha\beta} - 2\alpha\beta$

$\Rightarrow \frac{5}{4} - 2(4)$

$\Rightarrow \frac{5}{4} - 8 = \frac{5-32}{4} = -\frac{27}{4}$

Evaluate

(ii) $\alpha - \beta = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$

$= \sqrt{5^2 - 4(4)}$

$= \sqrt{25-16} = \sqrt{9} = 3$

$= \sqrt{\alpha^2+\beta^2+2\alpha\beta-4\alpha\beta}$

$= \sqrt{\alpha^2+\beta^2-2\alpha\beta}$

$= \sqrt{(\alpha-\beta)^2}$

1.	**Question:** Given that $\mathrm{HCF}(336, 54) = 6$, what is $\mathrm{LCM}(336, 54)$? **(A)** 18144 **(B)** 3024 **(C)** 56 **(D)** 9	1
2.	**Question:** For the linear polynomial $p(x) = mx + \frac{1}{n}$, if the sum of the zero of the polynomial and the constant term $\frac{1}{n}$ is 1, which of the following expressions represents the coefficient m in terms of n? **(A)** $m = \frac{1}{1-n}$ **(B)** $m = \frac{n+1}{n}$ **(C)** m =1− n **(D)** $m = \frac{1-n}{n}$	1

3.	**Question:** What is the HCF and LCM of the numbers 144 and 198 using prime factorisation? **(A)** $\text{HCF} = 18, \text{LCM} = 1584$ **(B)** $\text{HCF} = 6, \text{LCM} = 1584$ **(C)** $\text{HCF} = 12, \text{LCM} = 1584$ **(D)** $\text{HCF} = 18, \text{LCM} = 792$	1
4.	**Question:** Given that the LCM of two numbers is 2079 and their HCF is 27, if one of the numbers is 297, what is the other number? **(A)** 77 **(B)** 189 **(C)** 7 **(D)** 56133	1
5.	**Question:** If α and β are the zeroes of the polynomial $x^2 - 5x + 7$, what is the polynomial whose zeroes are $1 + \frac{1}{\alpha}$ and $1 + \frac{1}{\beta}$?	1

Polynomial $= k[x^2 - (\text{sum of zeroes})x + \text{product of zeroes}]$

$\alpha + \beta = 5$

$\alpha\beta = 7$

Sum of zeroes $= 1 + \frac{1}{\alpha} + 1 + \frac{1}{\beta}$

$= 2 + \frac{\beta + \alpha}{\alpha\beta}$

$= 2 + \frac{5}{7}$

$= \frac{19}{7}$

Product of zeroes $= \left(1 + \frac{1}{\alpha}\right)\left(1 + \frac{1}{\beta}\right)$

$= 1 + \frac{1}{\beta} + \frac{1}{\alpha} + \frac{1}{\alpha\beta}$

$= 1 + \frac{\alpha + \beta}{\alpha\beta} + \frac{1}{\alpha\beta}$

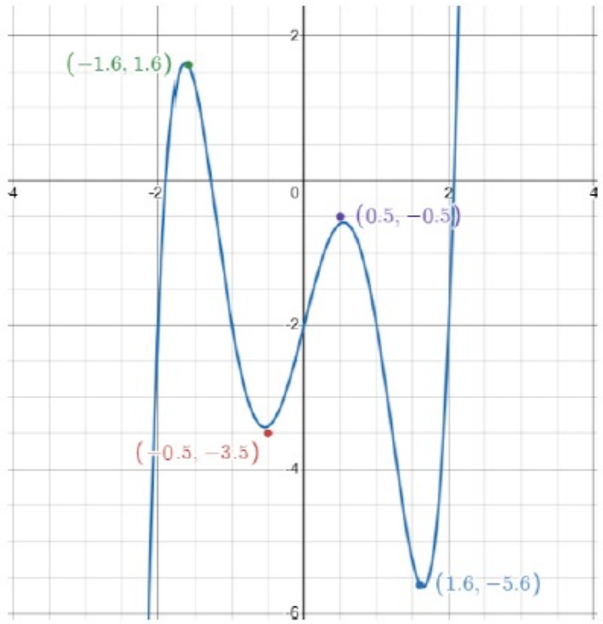
$= 1 + \frac{5}{7} + \frac{1}{7}$

$= 1 + \frac{6}{7}$

$= \frac{13}{7}$

$7k\left[x^2 - \frac{19}{7}x + \frac{13}{7}\right]$

$7x^2 - 19x + 13$

6.	**Question:** Consider the below graph of the corresponding equation of a polynomial p(x).  If the entire graph is translated 2 units upward, resulting in a new polynomial p′(x), how many real zeroes does p′(x) have? **(A)** 3 **(B)** 4 **(C)** 5 **(D)** 6	1